

DEMAND FORECASTING USING MACHINE LEARNING

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Abstract

Demand forecasting predicts future customer demand using historical sales data and machine learning. It helps optimize inventory, reduce costs, and improve customer satisfaction.

Introduction

Demand forecasting supports inventory planning, supply chain management, and production scheduling. Machine learning captures complex sales patterns more effectively than traditional methods.

Theory

The system uses supervised learning with preprocessing, feature engineering, model training, prediction, and evaluation. Algorithms include Linear Regression, Random Forest, Decision Tree, Gradient Boosting, and XGBoost.

Objectives

- Collect and preprocess data.
- Build a demand prediction model.
- Improve forecasting accuracy.
- Reduce shortages and overstock.
- Evaluate using MAE, RMSE, and R^2 .

Methodology

1. Data Collection
2. Data Preprocessing
3. Feature Engineering
4. Model Training
5. Prediction
6. Evaluation

Dataset

Product ID, Product Category, Date, Historical Sales, Price, Promotion, Holiday Indicator, Weather (optional), Region.

Mathematical Model

$$Y = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \dots + \beta_n X_n + \varepsilon$$

Complexity Analysis

Preprocessing: $O(n)$

Training: $O(n \times m)$

Prediction: $O(m)$

Memory: $O(n)$

Results

Accurate demand prediction, better inventory planning, reduced shortages, minimized excess stock, and improved business decisions.

Applications

Retail, E-Commerce, Supply Chain, Manufacturing, Warehousing, Pharmaceuticals, Food & Beverage, Sales Planning.

Conclusion

The model accurately forecasts demand and helps organizations improve operational efficiency and decision-making.